# Well Intelligence: evaluation report

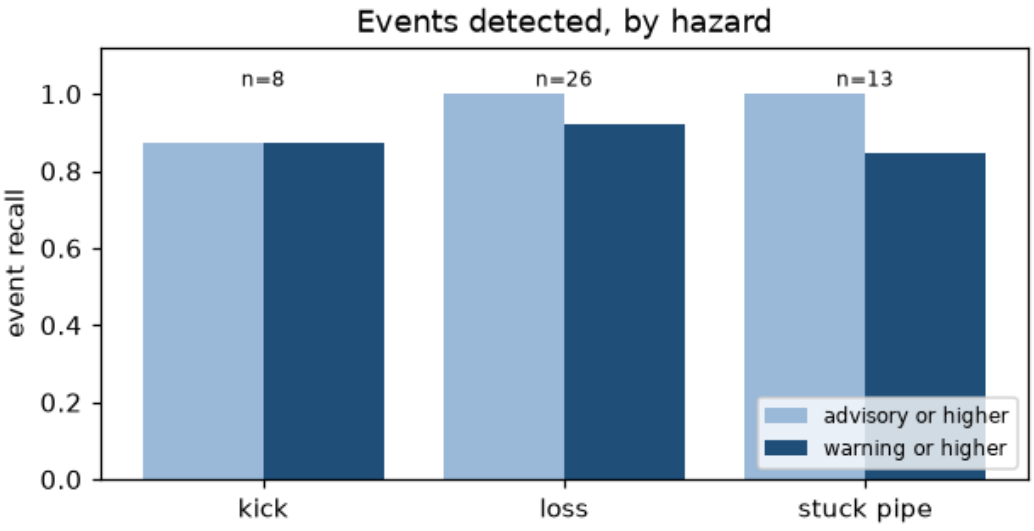
2026-10-08 00:52 UTC · python -m api.replay evaluate · detectors from the saved run

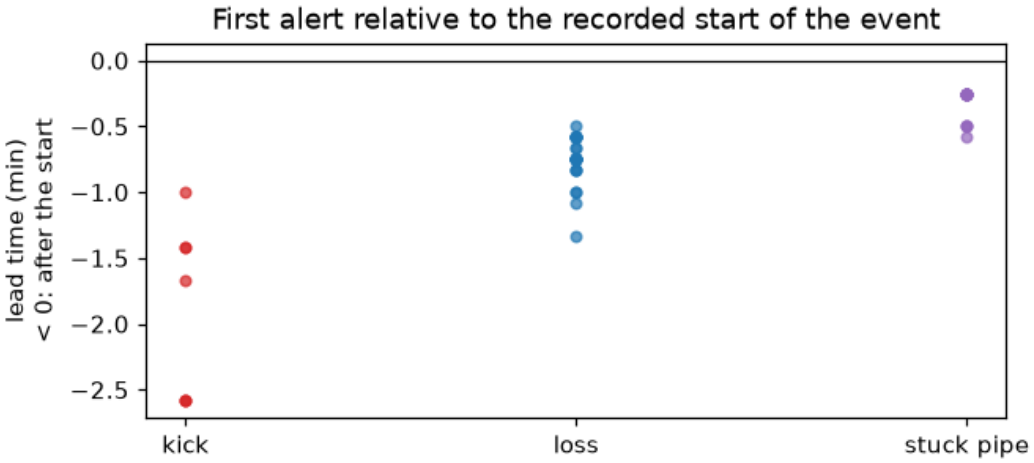
All data synthetic (synthetic_assam): an upper bound, not field performance.

| Measure | Value |
|---|---|
| Alert event recall (warning or higher) | **89%** of 47 recorded events (98% advisory or higher) |
| Median lead time (first alert vs the event's recorded start) | **-0.7 min** (0 of 46 detected events warned before they started) |
| Decision window (first alert to the event's recorded end) | **19.0 min** median over 46 detected events |
| False alerts per 24 operating hours | **0.09** (2 in 561 h) |
| Brier score (all hazards, every evaluation) | **0.00040**, skill 0.92 vs always predicting the base rate |
| Extraction, typed documents | precision **1.000**, recall **1.000** |
| Extraction, scanned documents | precision **0.966**, recall **0.850** |
| Extraction, fax documents | precision **0.405**, recall **0.043** |
| Extraction, handwritten documents | precision **0.722**, recall **0.368** |
| Assistant answer accuracy (rules) | **100%** of 100 questions, citations 100%, verifier reject rate 0.0% |
| Assistant with the configured LLM | not measured in this run (--chat-llm) |

## Alerts by hazard (leave-one-well-out replay)

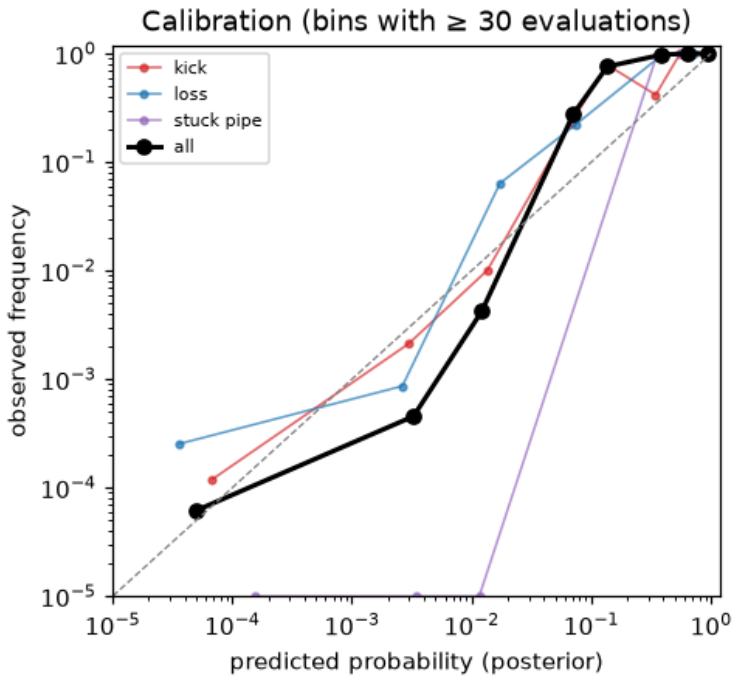
| Hazard | Events | Detected (advisory+) | Detected (warning+) | Median lead (min) | Before start | False alerts | per 24 h |
|---|---|---|---|---|---|---|---|
| kick | 8 | 7 (88%) | 7 (88%) | -1.7 | 0 | 2 | 0.09 |
| loss | 26 | 26 (100%) | 24 (92%) | -0.8 | 0 | 0 | 0.00 |
| stuck pipe | 13 | 13 (100%) | 11 (85%) | -0.2 | 0 | 0 | 0.00 |



# Calibration

| Hazard | Evaluations | Base rate | Brier | Brier (base rate only) | Skill |
|---|---|---|---|---|---|
| kick | 409,490 | 0.0035 | 0.00074 | 0.00344 | 0.78 |
| loss | 409,490 | 0.0193 | 0.00111 | 0.01891 | 0.94 |
| pack off | 409,490 | 0.0000 | 0.00000 | 0.00000 | – |
| stuck pipe | 409,490 | 0.0063 | 0.00054 | 0.00630 | 0.91 |
| tight hole | 409,490 | 0.0000 | 0.00000 | 0.00000 | – |
| washout | 409,490 | 0.0000 | 0.00000 | 0.00000 | – |
| all | 2,456,940 | 0.0048 | 0.00040 | 0.00482 | 0.92 |



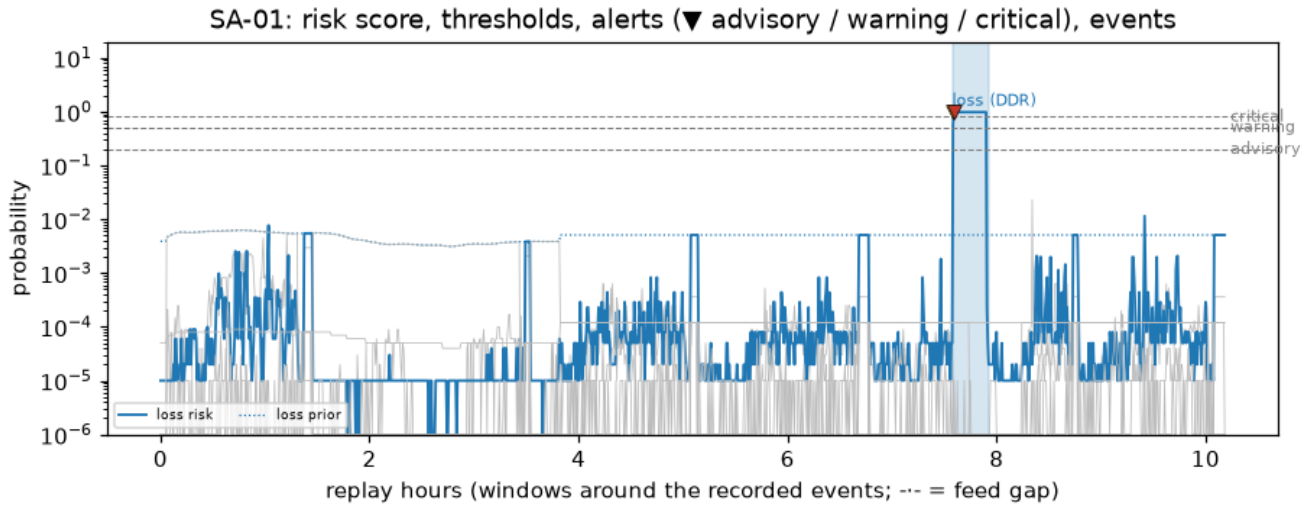
## Per well

| Well | Offsets (leave-one-out) | Events | Detected (warning+) | Lead (min) | Decision window (min) | False alerts | Operating h |
|---|---|---|---|---|---|---|---|
| SA-01 | SA-11, SA-30, SA-14, SA-08 … | loss | 1/1 | -0.8 | 19.0 | 0 | 10.2 |
| SA-02 | SA-24, SA-15, SA-29, SA-01 … | none | 0/0 | – | – | 0 | 14.5 |
| SA-03 | SA-04, SA-24, SA-22, SA-29 … | loss, loss | 2/2 | -0.8, -0.6 | 19.0, 31.0 | 0 | 20.8 |
| SA-04 | SA-22, SA-03, SA-29, SA-18 … | kick, loss | 2/2 | -2.6, -0.6 | 12.3, 31.0 | 0 | 20.5 |
| SA-05 | SA-19, SA-06, SA-09, SA-18 | loss | 1/1 | -0.8 | 19.0 | 0 | 10.2 |
| SA-06 | SA-19, SA-18, SA-05, SA-11 … | loss | 1/1 | -0.8 | 19.0 | 0 | 10.2 |
| SA-07 | SA-13, SA-17, SA-26, SA-22 … | none | 0/0 | – | – | 0 | 14.5 |
| SA-08 | SA-14, SA-25, SA-11, SA-01 | stuck pipe, loss, stuck pipe | 3/3 | -0.2, -0.6, -0.2 | 12.0, 31.0, 26.4 | 0 | 32.3 |
| SA-09 | SA-19, SA-05, SA-06, SA-30 … | stuck pipe | 1/1 | -0.2 | 12.0 | 0 | 10.3 |
| SA-10 | SA-21, SA-20, SA-16, SA-17 … | loss, stuck pipe | 2/2 | -0.8, -0.2 | 19.0, 21.0 | 0 | 20.5 |
| SA-11 | SA-30, SA-18, SA-01, SA-06 … | loss | 1/1 | -0.8 | 18.9 | 0 | 10.2 |
| SA-12 | SA-22, SA-13, SA-15, SA-07 … | none | 0/0 | – | – | 0 | 14.5 |
| SA-13 | SA-07, SA-12, SA-29, SA-15 … | kick | 1/1 | -2.6 | 12.3 | 0 | 9.9 |
| SA-14 | SA-08, SA-25, SA-11, SA-01 | stuck pipe, loss, loss | 2/3 | -0.6, -0.6, -1.0 | 11.7, 31.0, 23.6 | 0 | 32.1 |
| SA-15 | SA-29, SA-24, SA-12, SA-22 … | none | 0/0 | – | – | 0 | 14.5 |
| SA-16 | SA-27, SA-20, SA-10, SA-21 … | loss | 1/1 | -0.8 | 19.0 | 0 | 10.2 |
| SA-17 | SA-18, SA-21, SA-07, SA-10 … | kick, loss, stuck pipe | 2/3 | -0.7, -0.2 | 30.9, 26.4 | 0 | 31.9 |
| SA-18 | SA-17, SA-11, SA-30, SA-06 … | none | 0/0 | – | – | 0 | 14.5 |
| SA-19 | SA-06, SA-24, SA-09, SA-05 … | stuck pipe | 1/1 | -0.5 | 11.8 | 0 | 10.3 |
| SA-20 | SA-10, SA-21, SA-26, SA-16 … | loss, kick | 2/2 | -0.8, -1.4 | 19.0, 13.5 | 0 | 19.9 |
| SA-21 | SA-27, SA-17, SA-10, SA-20 … | loss, loss | 2/2 | -0.8, -0.6 | 19.0, 31.0 | 0 | 20.8 |
| SA-22 | SA-04, SA-12, SA-03, SA-15 … | loss, loss, loss | 2/3 | -1.1, -0.6, -0.8 | 18.7, 31.0, 23.8 | 0 | 32.0 |
| SA-23 | SA-28 | stuck pipe, kick, loss | 3/3 | -0.2, -1.0, -0.8 | 12.0, 13.9, 19.2 | 0 | 24.3 |

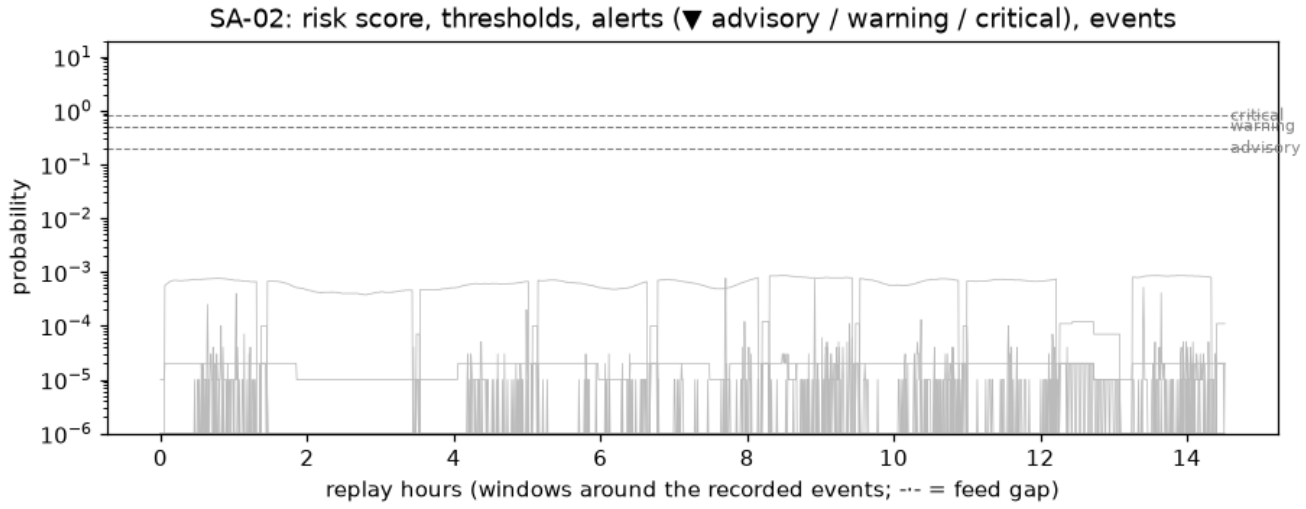
| Well | Offsets (leave-one-out) | Events | Detected (warning+) | Lead (min) | Decision window (min) | False alerts | Operating h |
|---|---|---|---|---|---|---|---|
| SA-24 | SA-29, SA-15, SA-03, SA-19 … | kick | 1/1 | -2.6 | 12.3 | 0 | 9.9 |
| SA-25 | SA-14, SA-08, SA-30, SA-01 | stuck pipe, kick | 2/2 | -0.2, -1.4 | 12.0, 13.5 | 0 | 20.0 |
| SA-26 | SA-07, SA-20, SA-16, SA-13 … | loss | 1/1 | -1.0 | 18.8 | 0 | 10.2 |
| SA-27 | SA-21, SA-16, SA-10, SA-20 | none | 0/0 | – | – | 0 | 14.5 |
| SA-28 | SA-23 | loss, loss, kick, stuck pipe | 3/4 | -1.3, -0.6, -1.7, -0.2 | 18.4, 31.0, 11.2, 15.8 | 2 | 35.1 |
| SA-29 | SA-15, SA-24, SA-13, SA-04 … | stuck pipe, loss | 2/2 | -0.5, -0.8 | 11.8, 30.7 | 0 | 20.9 |
| SA-30 | SA-11, SA-18, SA-06, SA-19 … | stuck pipe, loss, stuck pipe, loss | 3/4 | -0.5, -0.7, -0.2, -0.5 | 11.8, 30.9, 26.4, 33.4 | 0 | 41.2 |

## Timelines

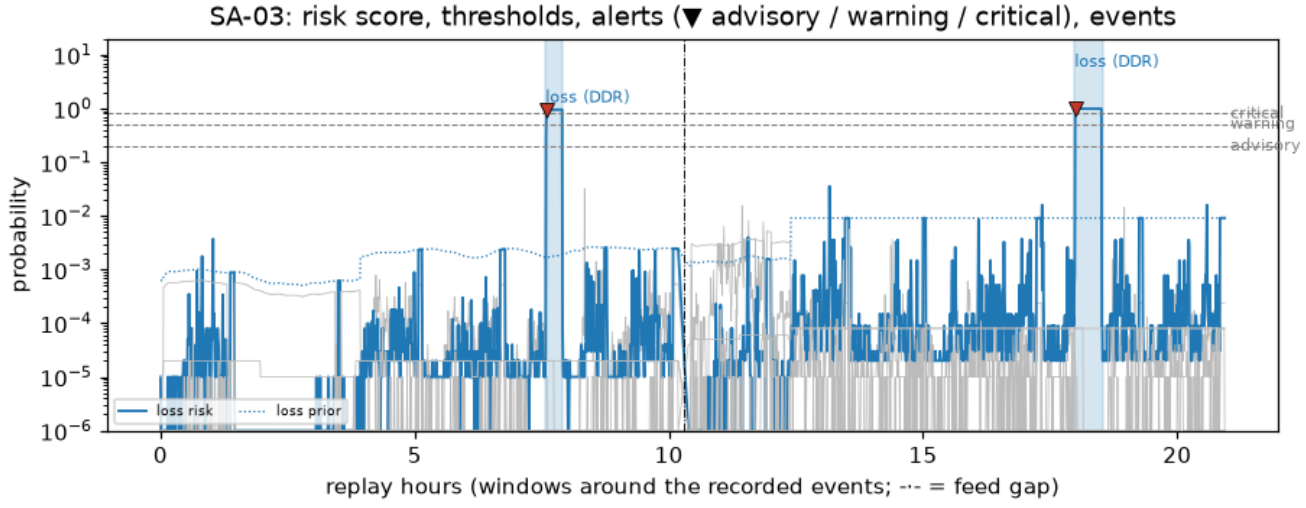
***SA-01***



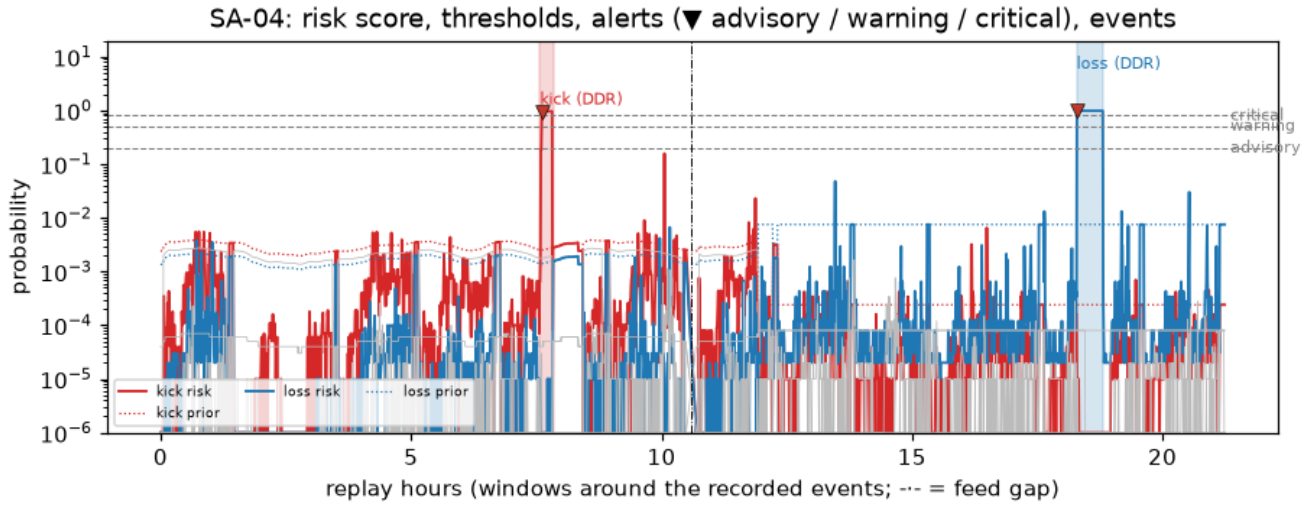

***SA-02***



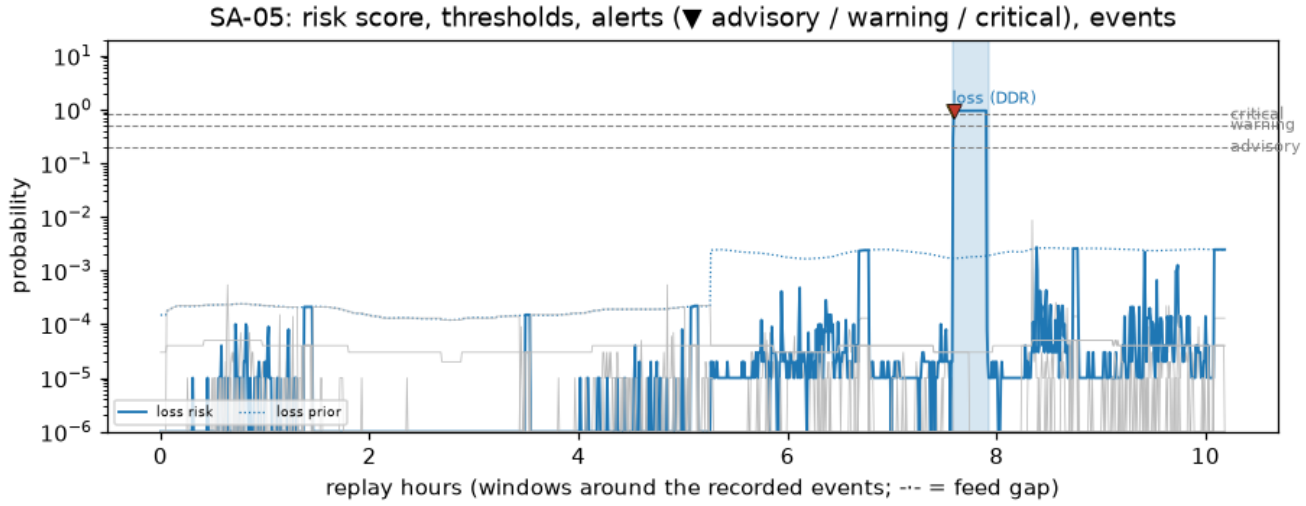

***SA-03***



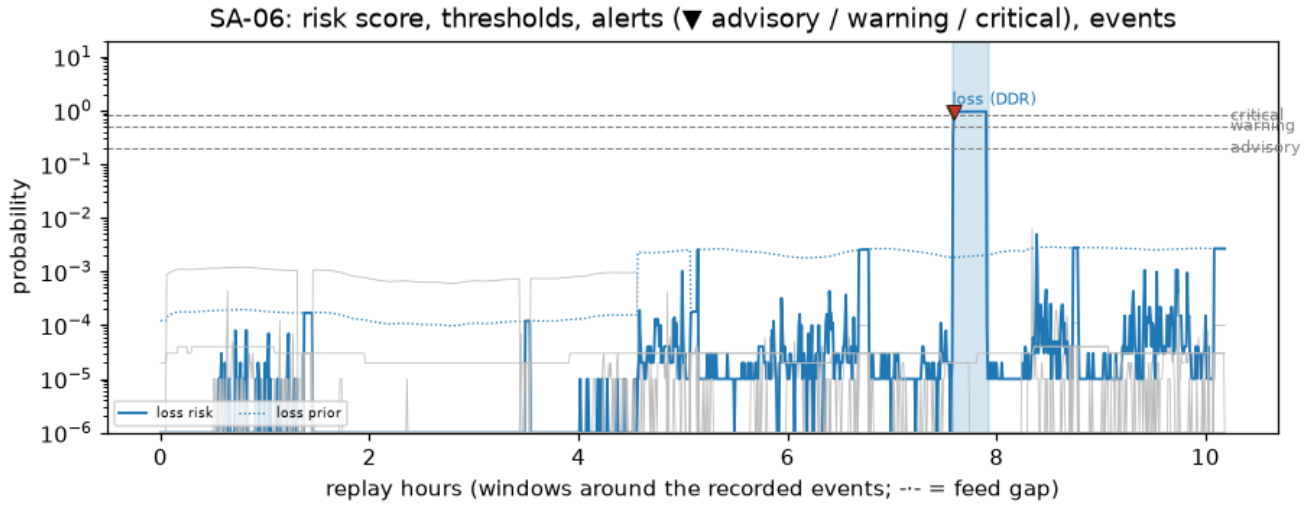

***SA-04***



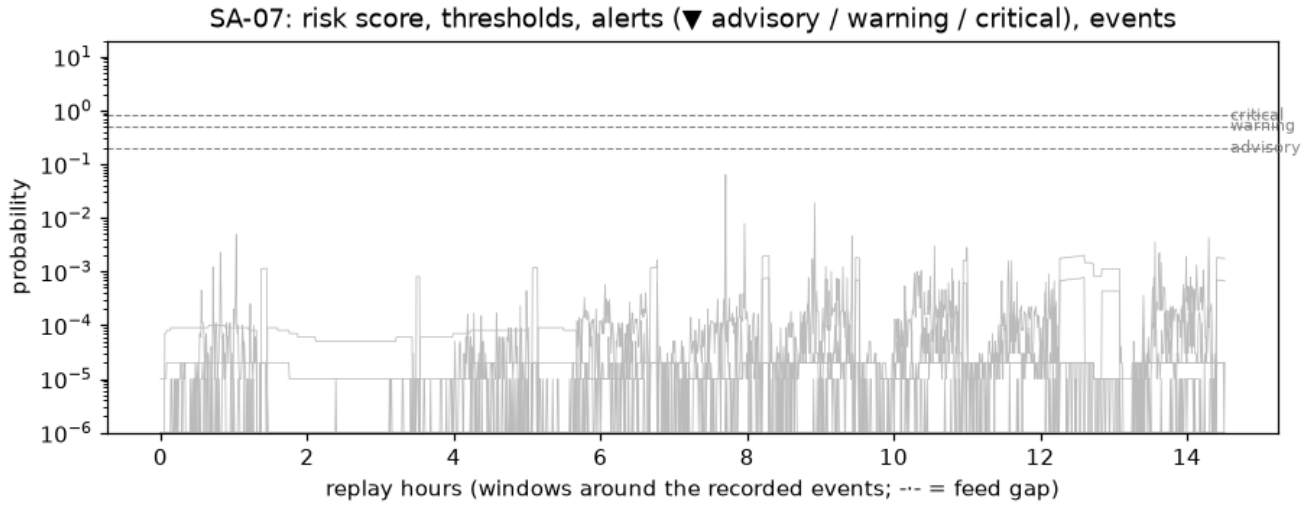

***SA-05***



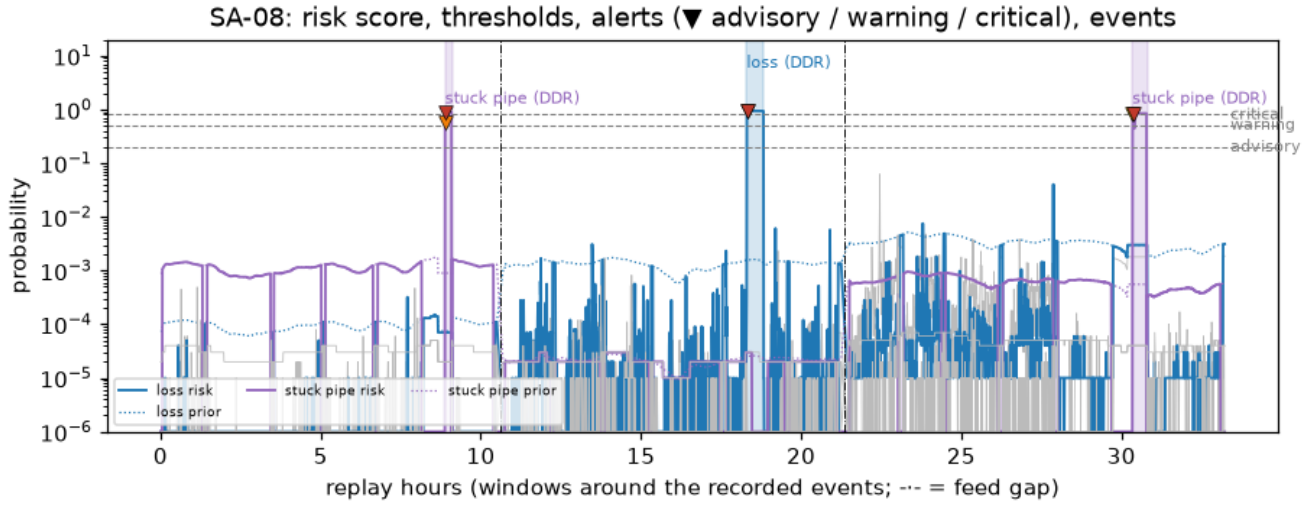

***SA-06***



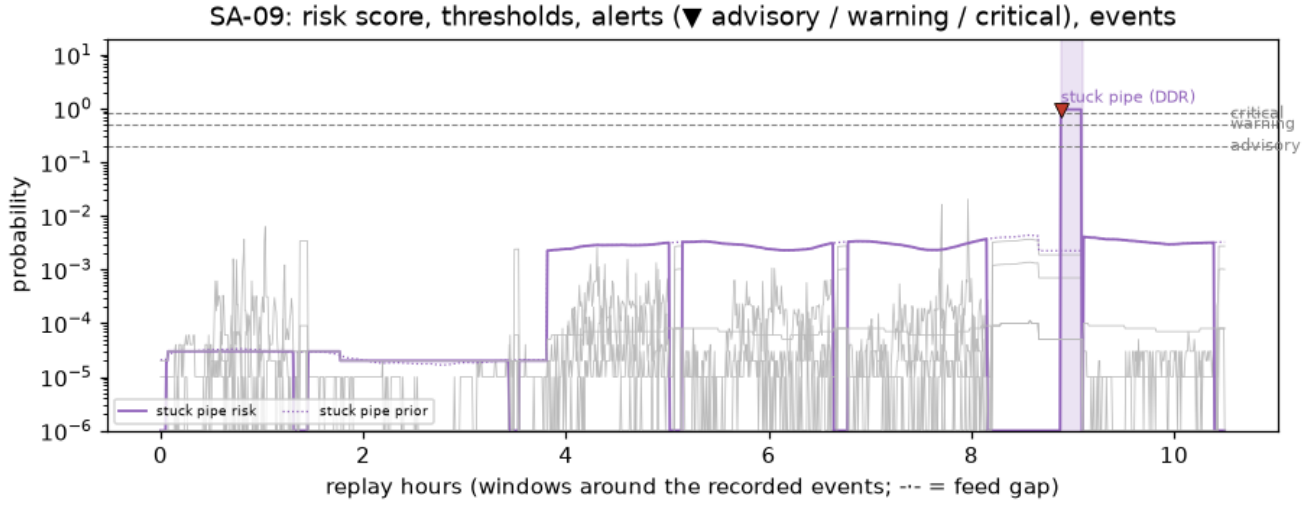

***SA-07***



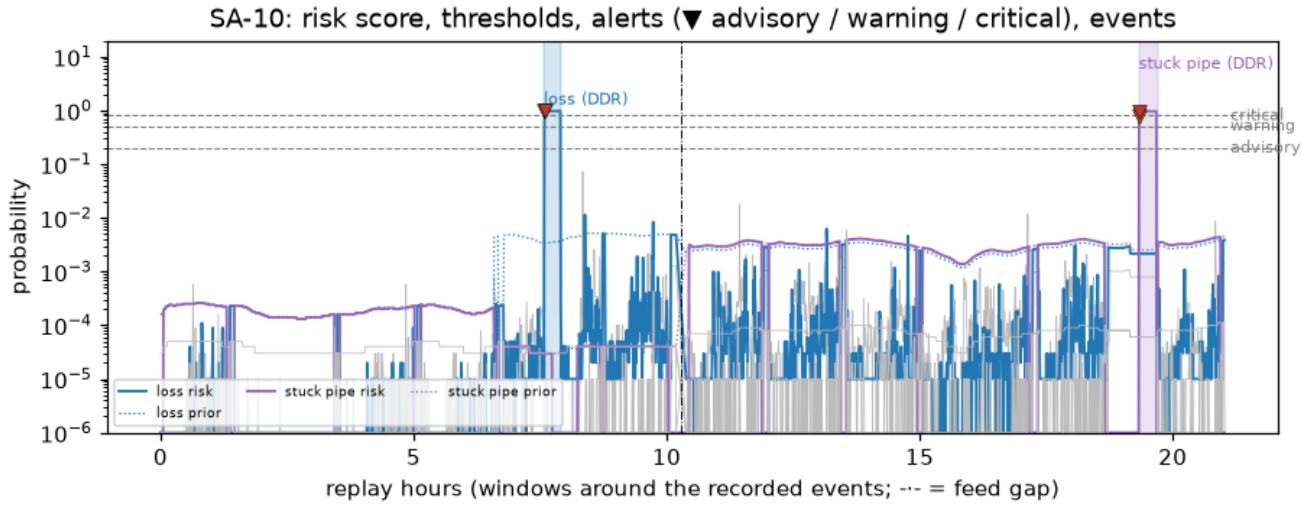

***SA-08***



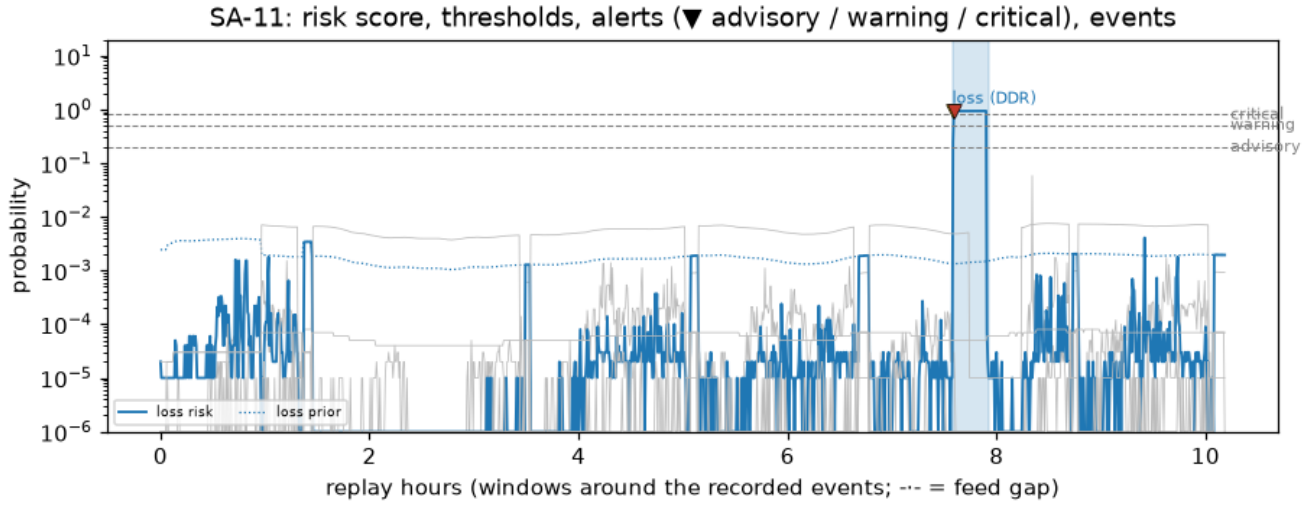

***SA-09***



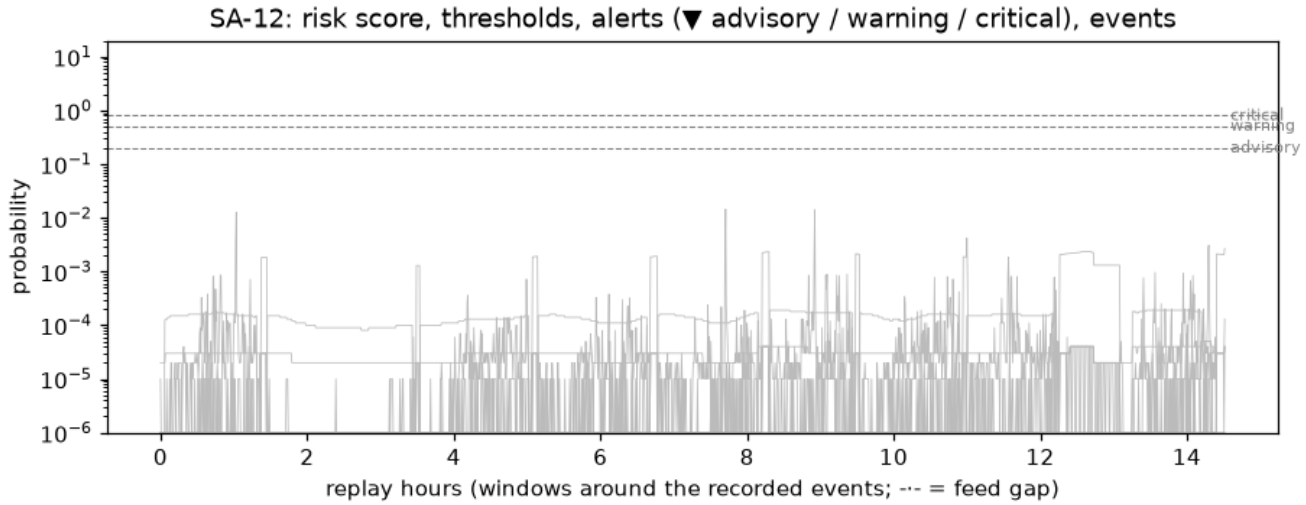

***SA-10***



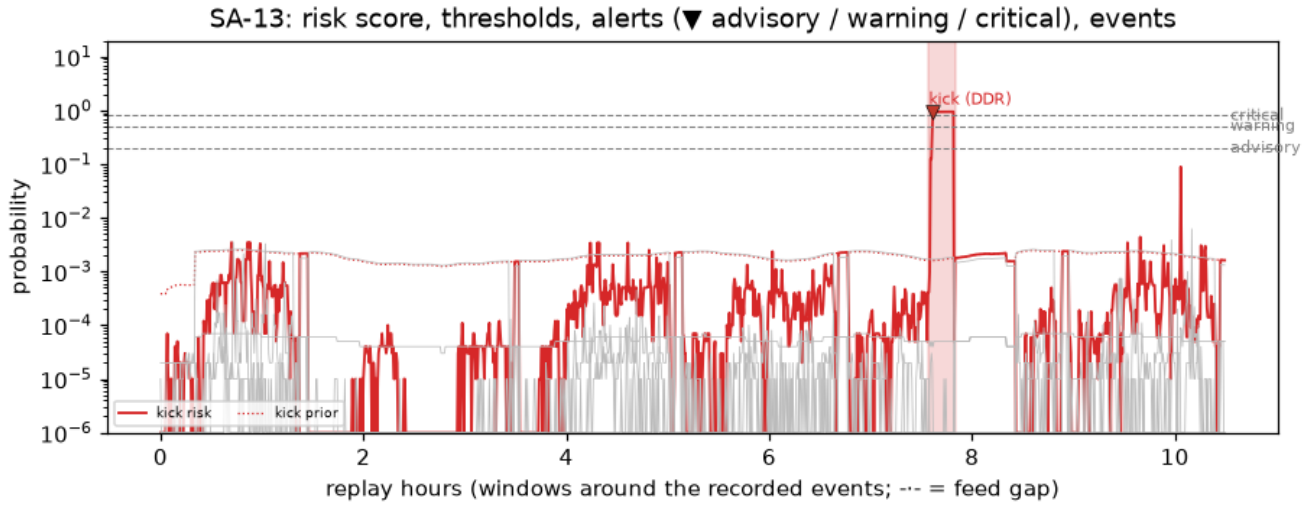

***SA-11***



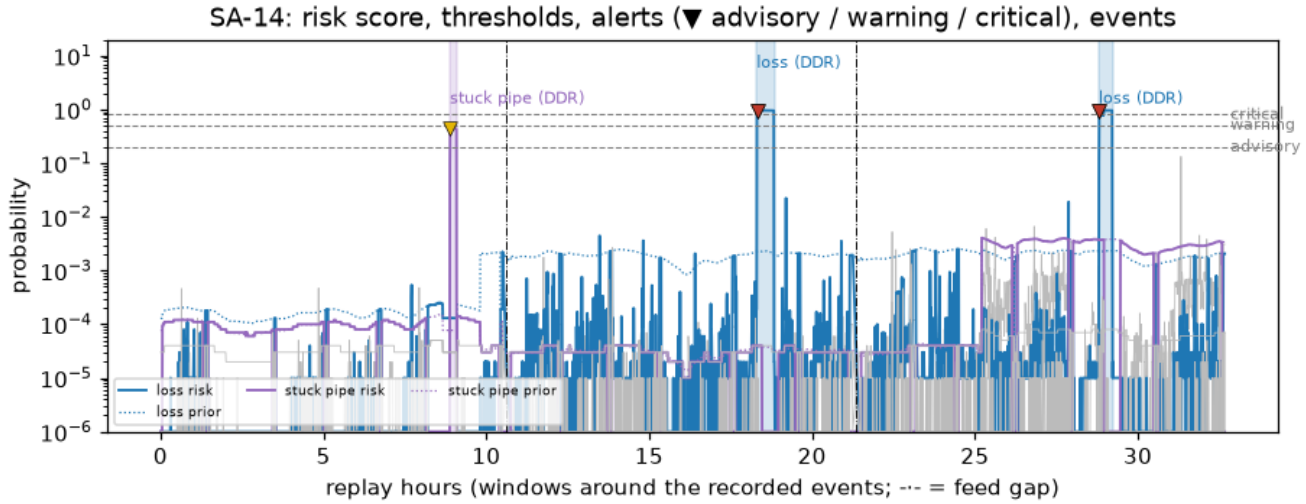

***SA-12***



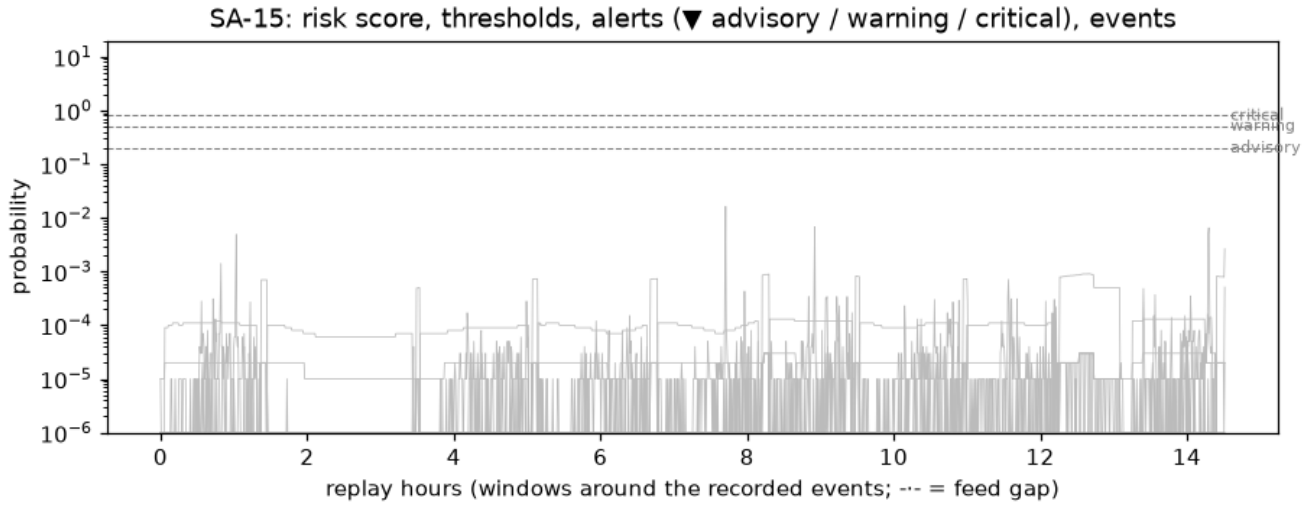

***SA-13***



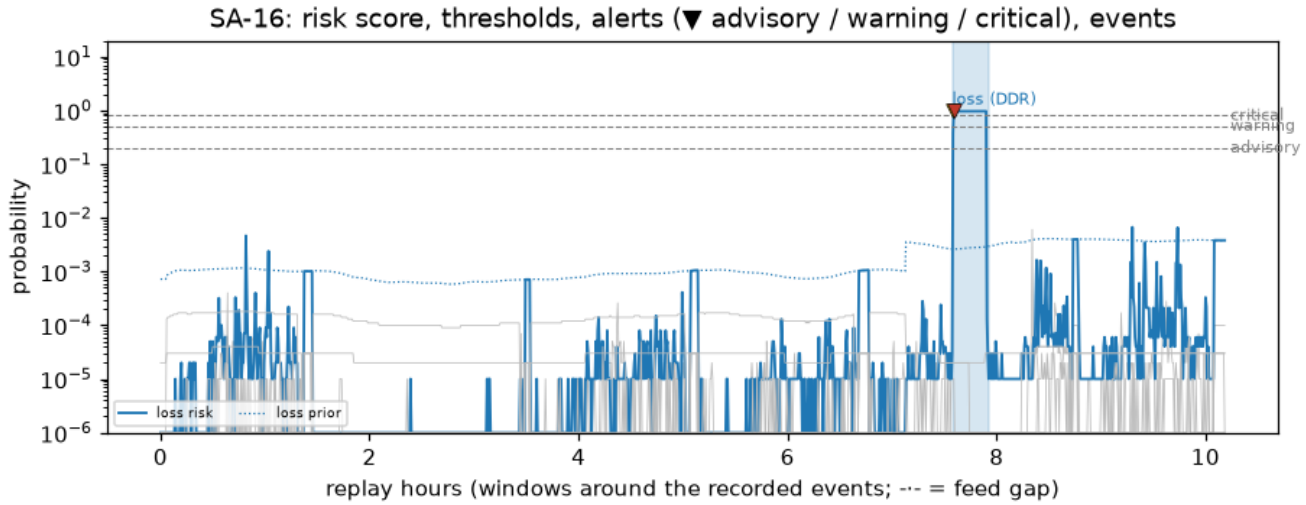

***SA-14***



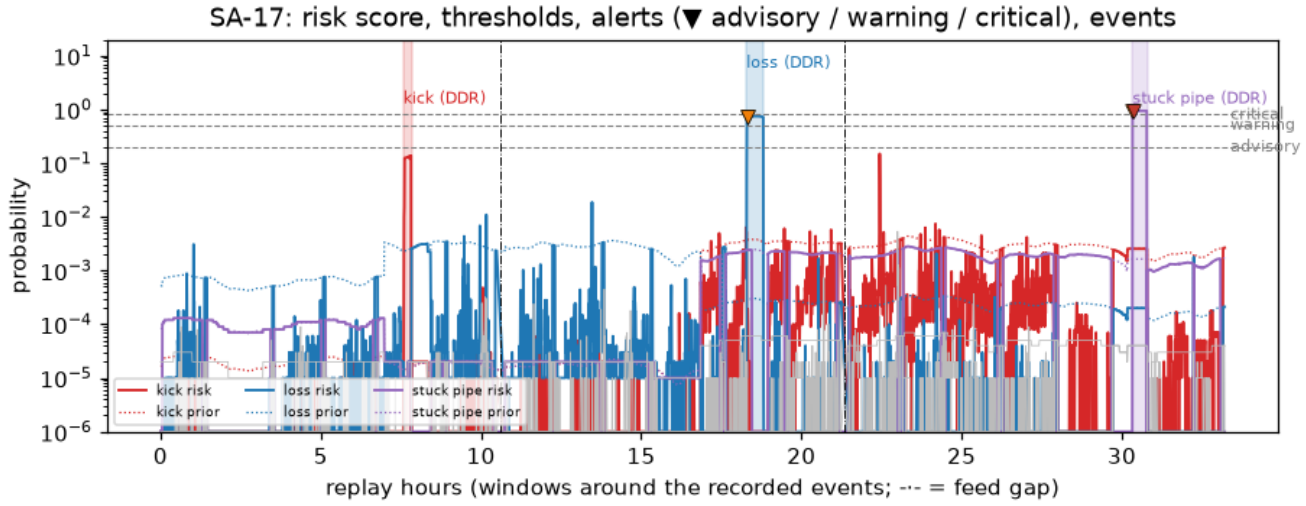

***SA-15***



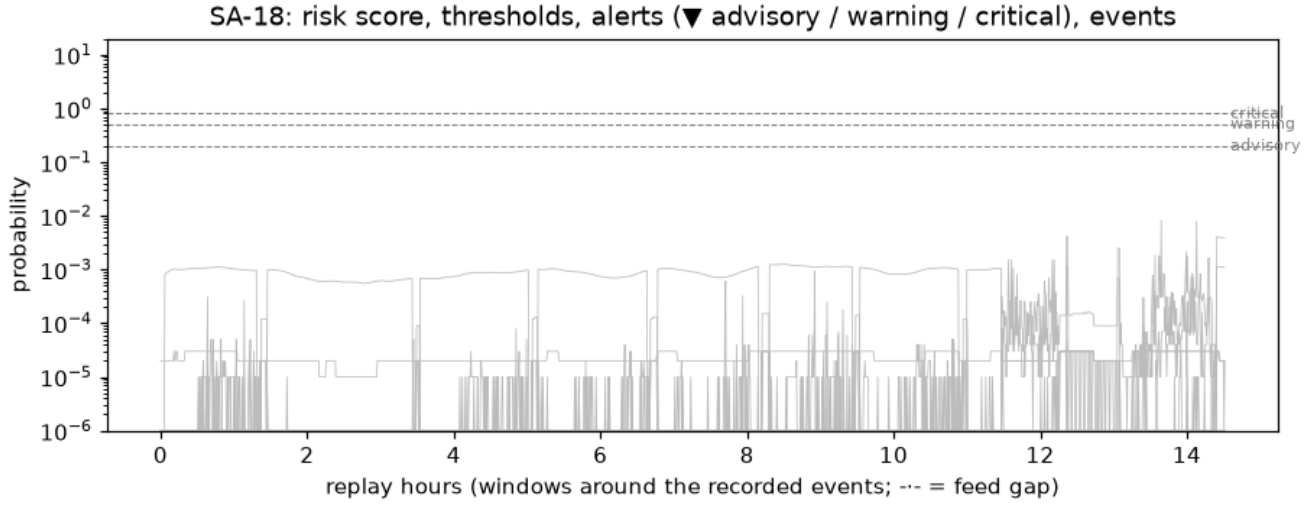

***SA-16***



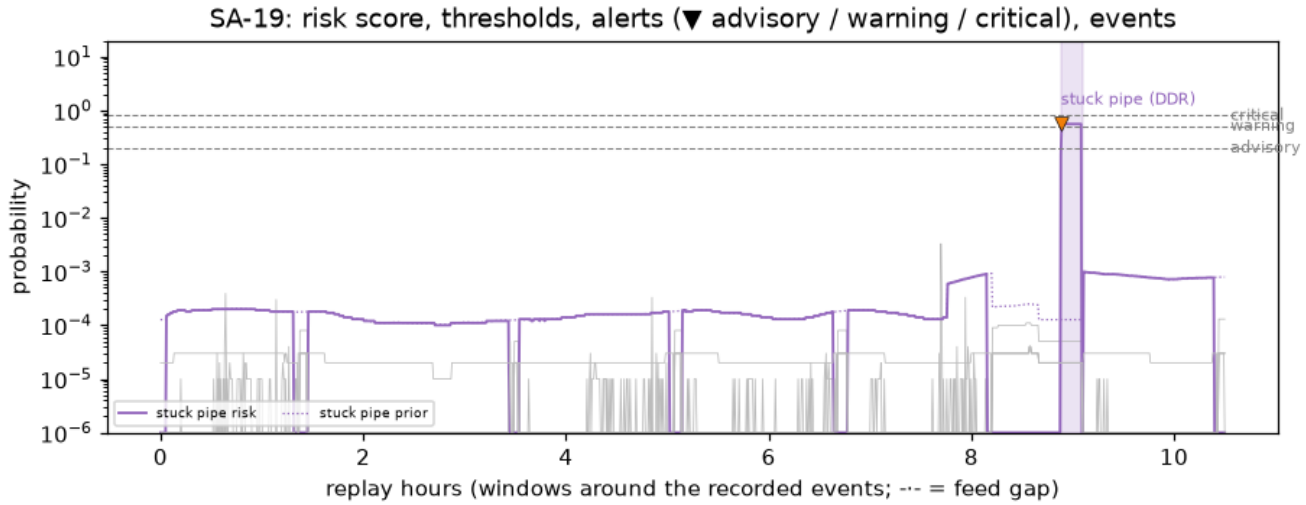

***SA-17***



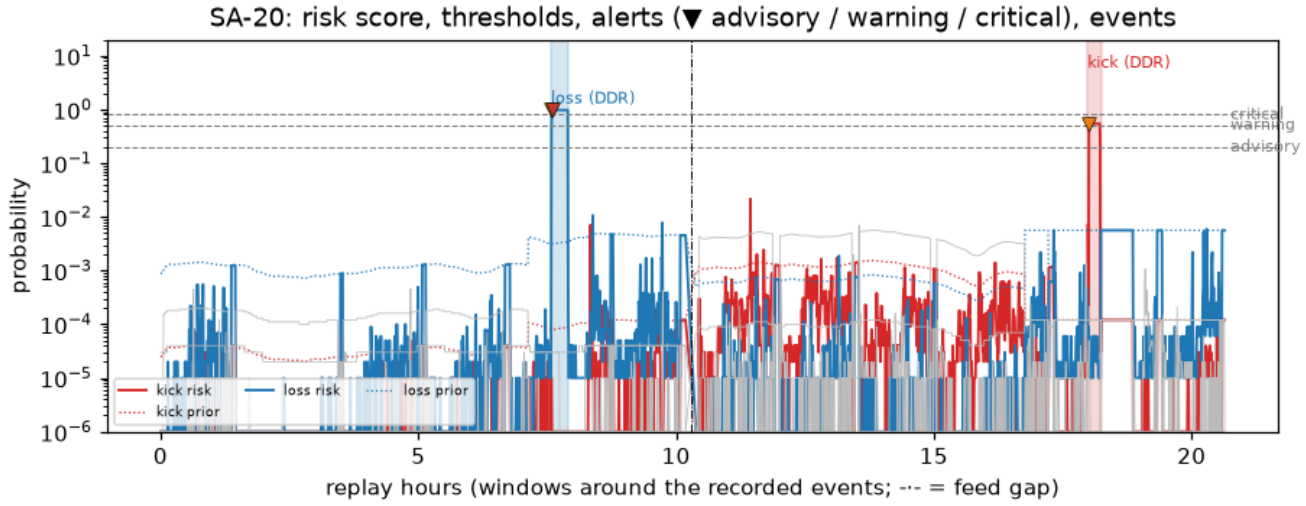

***SA-18***



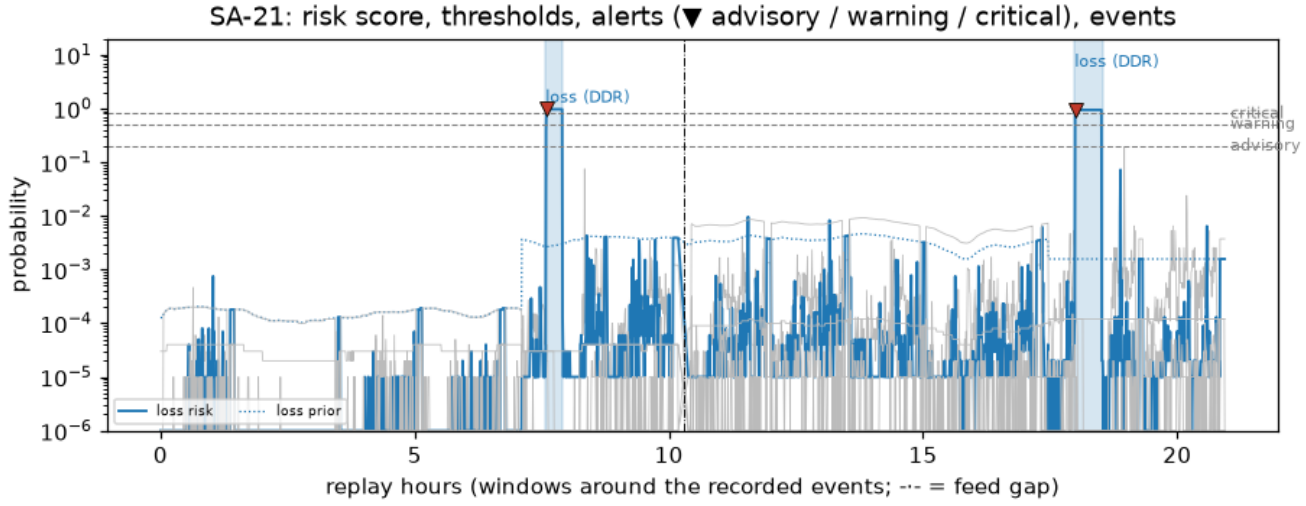

***SA-19***



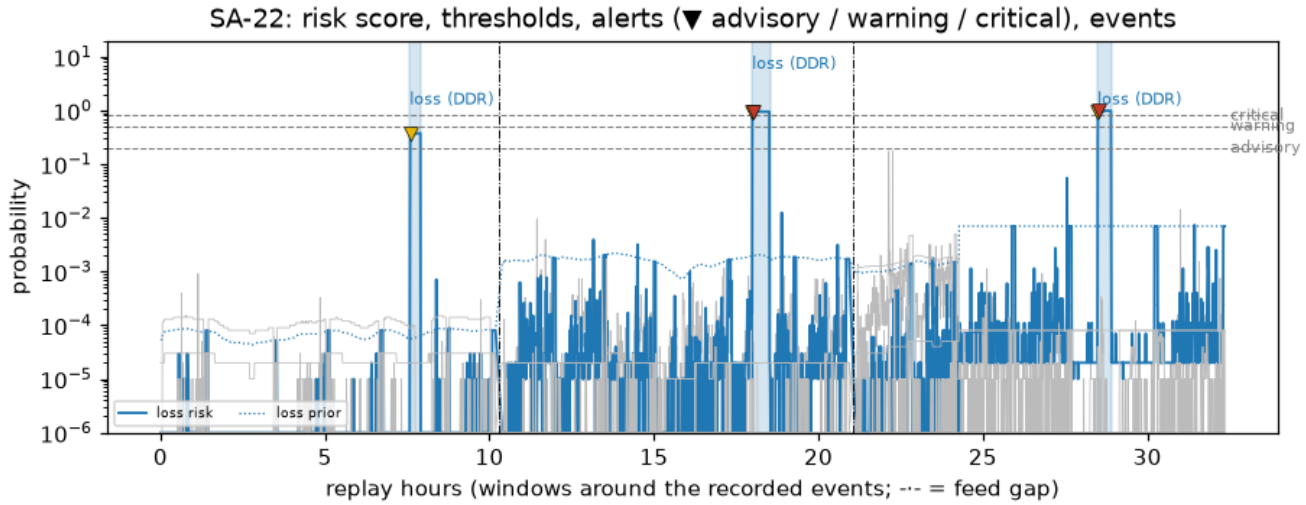

***SA-20***



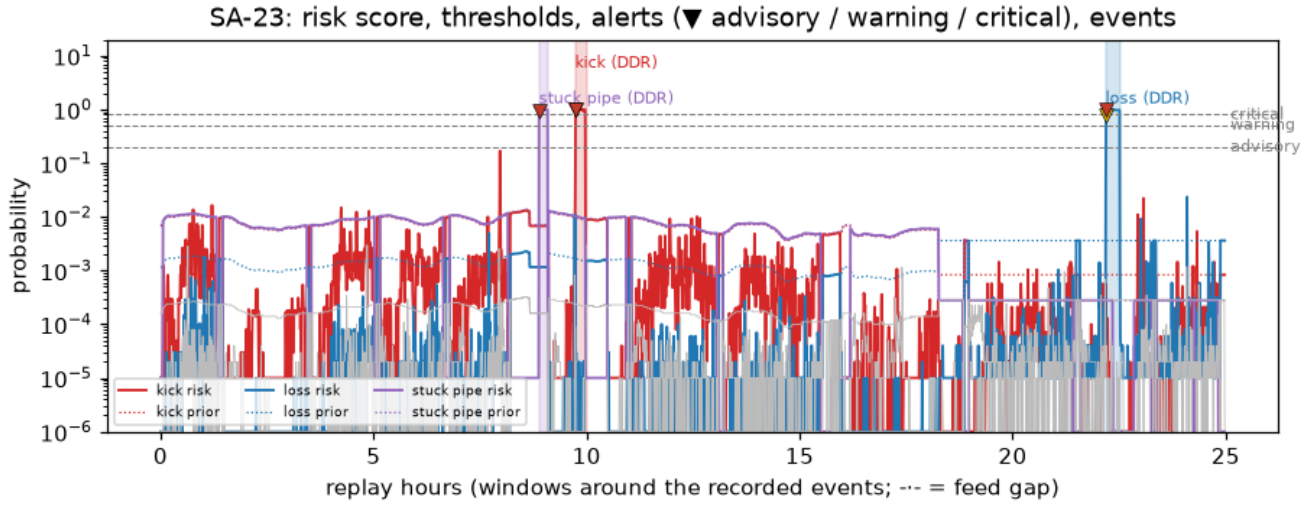

***SA-21***



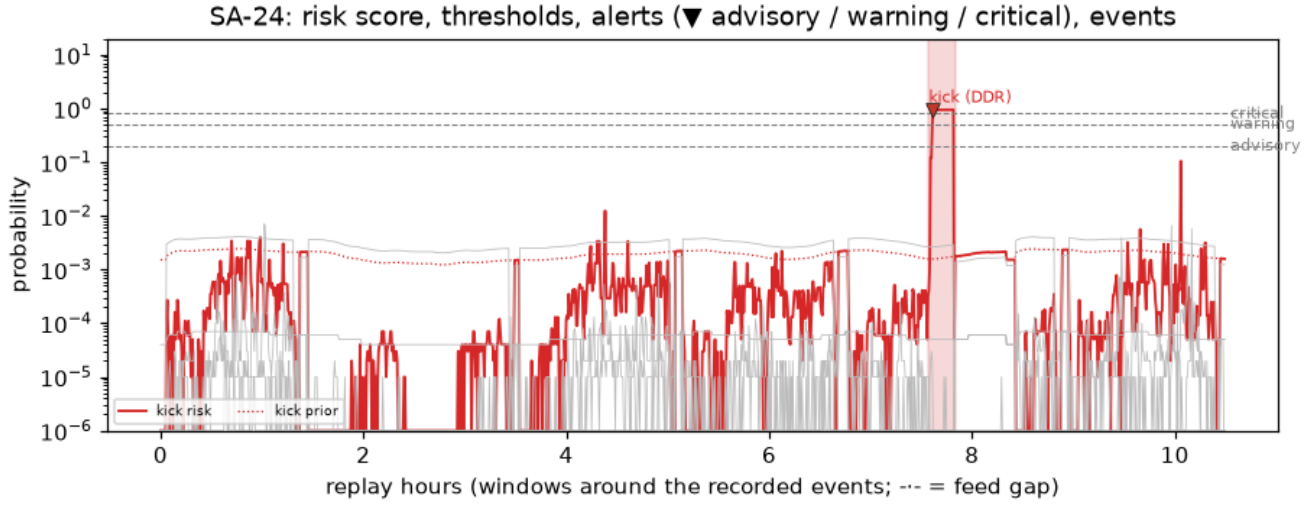

***SA-22***



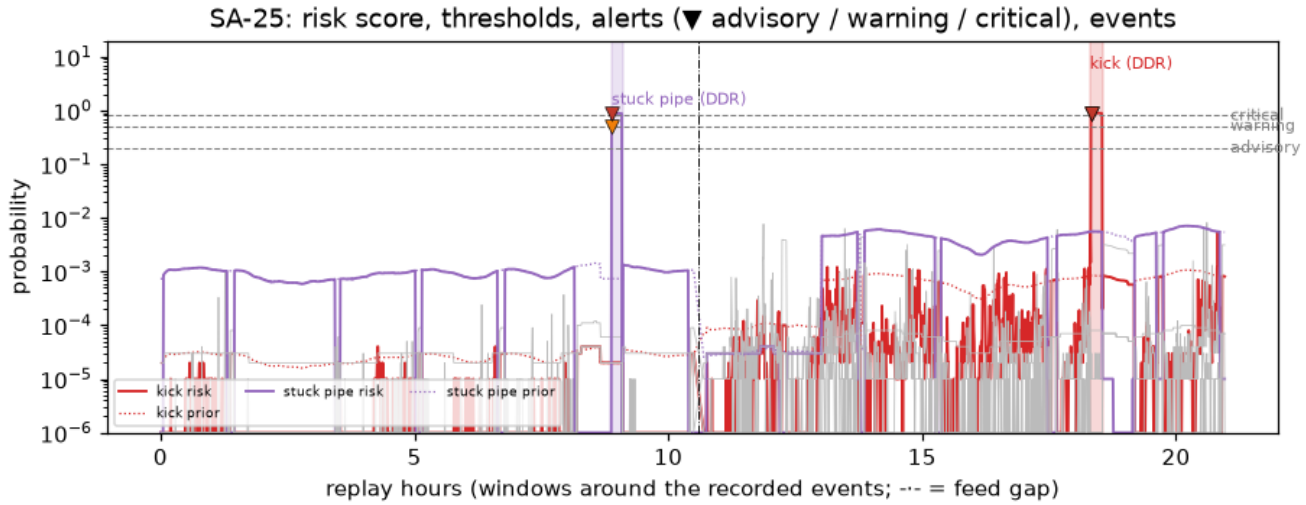

***SA-23***



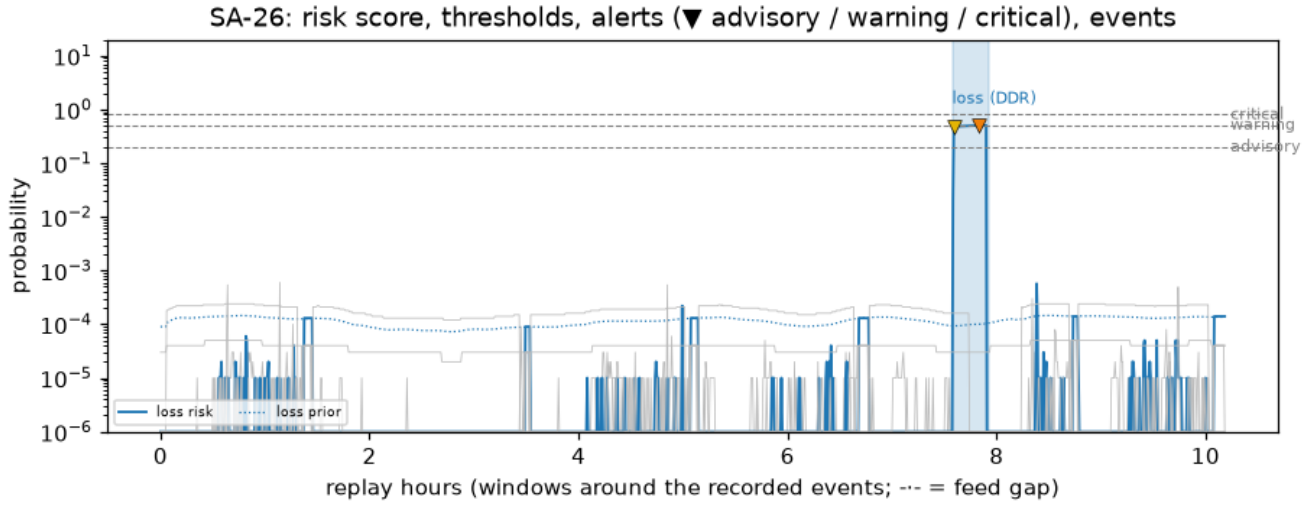

***SA-24***



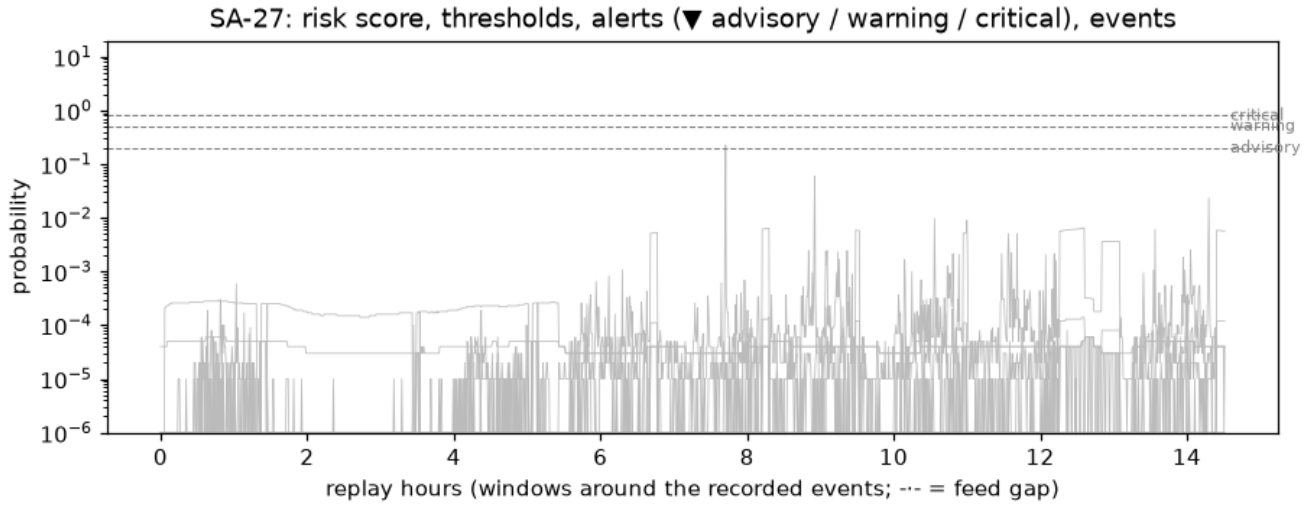

***SA-25***



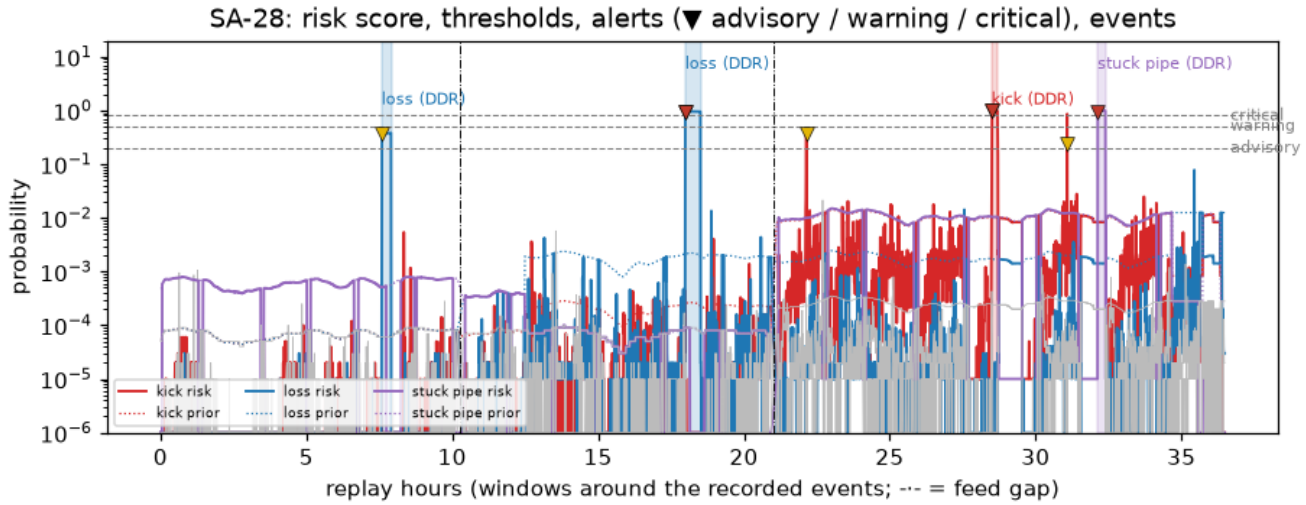

***SA-26***



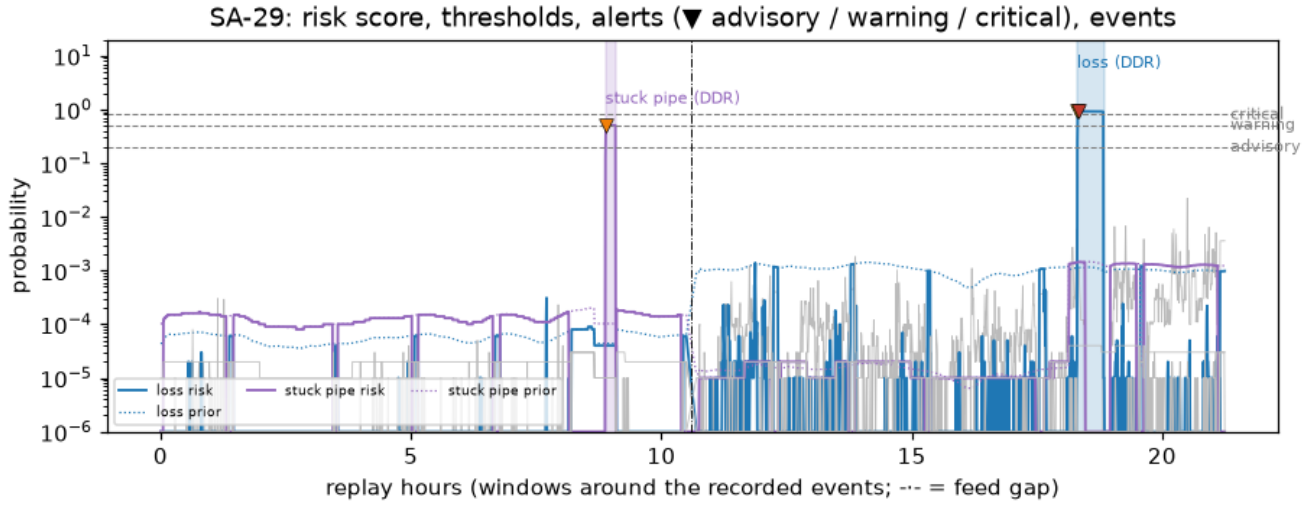

***SA-27***



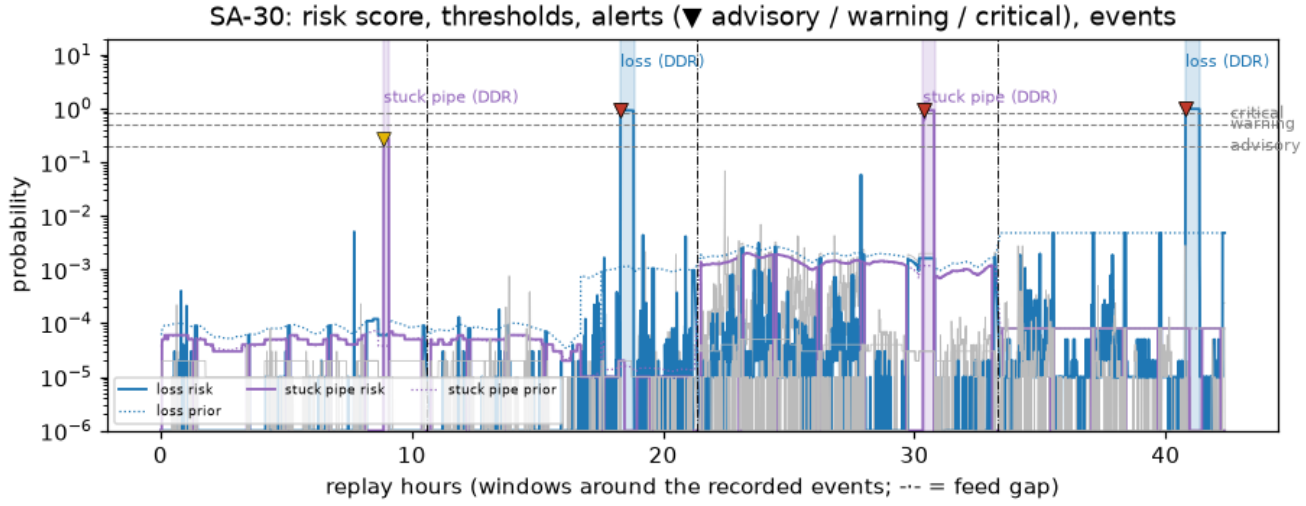

***SA-28***


SA-28: risk score, thresholds, alerts (▼ advisory / warning / critical), events
loss (DDR)
loss (DDR)
kick (DDR)
stuck pipe (DDR)
critical
warning
advisory
probability
kick risk
kick prior
loss risk
loss prior
stuck pipe risk
stuck pipe prior
replay hours (windows around the recorded events; --- = feed gap)

***SA-29***


SA-29: risk score, thresholds, alerts (▼ advisory / warning / critical), events
stuck pipe (DDR)
loss (DDR)
critical
warning
advisory
probability
loss risk
loss prior
stuck pipe risk
stuck pipe prior
replay hours (windows around the recorded events; --- = feed gap)

***SA-30***


SA-30: risk score, thresholds, alerts (▼ advisory / warning / critical), events
stuck pipe (DDR)
loss (DDR)
stuck pipe (DDR)
loss (DDR)
critical
warning
advisory
probability
loss risk
loss prior
stuck pipe risk
stuck pipe prior
replay hours (windows around the recorded events; --- = feed gap)